

Supplementary CVT Geometry Notes
Belt-Length Approximation Error and Geometric Ratio Sensitivity

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Scope

This note preserves two pieces of CVT geometry work that are useful outside the main CINDER manuscript but are not required to support its central transient mechanics argument. The first quantifies the effect of a common small-angle belt-length approximation using the current CINDER reference geometry. The second develops geometric-ratio rate and acceleration relations and uses them to visualize how belt length, centre distance, pulley radii, and sheave angle shape the available ratio change.

The note is intentionally tied to the reduced belt geometry used by CINDER. Where a result is inherited directly from the main formulation, the `\FormEq` and `\FormSec` helpers link to the corresponding equation or section in `../CVT_Module_Formulation/CVT_Module_Formulation.tex`.

Chapter 1

Small-Angle Belt-Length Approximation

The exact CINDER belt geometry follows the outer surface of the reduced belt path. For pulley outer-surface radii r_p and r_s and fixed centre-to-centre distance C , the exact path length is given in Equation 3.1.24. Written here for convenience,

$$L_{\text{geo}} = \pi(r_p + r_s) + 2(r_s - r_p) \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2}. \quad (1.1)$$

The installed geometry satisfies $L_{\text{geo}} = L_b$, as in Equation 3.1.25. A common reduction neglects the unequal-wrap correction when

$$\frac{|r_s - r_p|}{C} \ll 1.$$

The resulting relation is

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}. \quad (1.2)$$

This chapter updates the original approximation study using the current CINDER reference geometry and separates two different uses of Equation (1.2): inferring the installed centre distance from a measured reference position, and using the simplified relation after the true centre distance is already known.

1.1 Approximate Centre Distance

At a measured reference position with known r_p , r_s , and L_b , Equation (1.2) can be rearranged directly. Isolating the straight-span term and squaring gives

$$C_{\text{approx}} = \sqrt{(r_s - r_p)^2 + \frac{1}{4} [L_b - \pi(r_p + r_s)]^2}. \quad (1.3)$$

The current reference configuration uses the values in Table 1.1.

Quantity	Symbol	Value
Outer-path belt length	L_b	0.953 262 m
Belt radial thickness	h_b	0.015 570 2 m
Cord-layer depth	d_{cord}	0.002 54 m
Sheave half-angle	β	0.200 712 864 rad
Primary outer radius at zero shift	$r_{p,\text{min}}$	0.036 207 7 m
Secondary outer radius at zero shift	$r_{s,\text{max}}$	0.1016 m
Deadzone boundary	s_{dz}	0.002 489 2 m
Maximum shift coordinate	s_{max}	0.019 05 m

Table 1.1: Current reference geometry used for the approximation comparison.

Solving the exact compatibility relation for the centre distance gives

$$C_{\text{exact}} = 0.251\,616\,999\text{ m} = 9.9062\text{ in},$$

whereas Equation (1.3) gives

$$C_{\text{approx}} = 0.268\,255\,528\text{ m} = 10.5612\text{ in}.$$

The inferred centre distance is therefore high by

$$\Delta C = 0.016\,638\,529\text{ m} = 0.6551\text{ in}, \quad (1.4)$$

$$\frac{\Delta C}{C_{\text{exact}}} = 6.613\%. \quad (1.5)$$

At this reference position,

$$\frac{r_s - r_p}{C_{\text{exact}}} = 0.2599, \quad \delta = \sin^{-1}\left(\frac{r_s - r_p}{C_{\text{exact}}}\right) = 0.2629\text{ rad} \approx 15.06^\circ.$$

The omitted wrap-angle correction is therefore not especially small at the reference geometry used to infer C .

1.2 Propagation Along the Shift Path

After C has been inferred, the simplified relation can also be solved explicitly for the secondary radius. Collecting terms in r_s gives

$$r_{s,\text{approx}} = \frac{\pi L_b + (4 - \pi^2)r_p}{\pi^2 + 4} - \frac{2\sqrt{(\pi^2 + 4)C^2 - L_b^2 + 4\pi L_b r_p - 4\pi^2 r_p^2}}{\pi^2 + 4}. \quad (1.6)$$

The smaller physical root is used on the overdrive side. The primary outer radius follows the current piecewise radius map in Equation 3.1.16. At maximum shift,

$$r_p(s_{\text{max}}) = r_{p,\text{min}} + \frac{s_{\text{max}} - s_{\text{dz}}}{2 \tan \beta} = 0.076\,907\,166\text{ m}. \quad (1.7)$$

Two ratio definitions are useful for interpreting the geometry. The outer-surface geometric ratio used in the design maps is

$$R_g = \frac{r_s}{r_p}, \quad (1.8)$$

while the cord-line ratio relevant to the present no-slip kinematics uses $r_{j,\text{eff}} = r_j - d_{\text{cord}}$:

$$R_{\text{eff}} = \frac{r_s - d_{\text{cord}}}{r_p - d_{\text{cord}}}. \quad (1.9)$$

At zero shift, the approximate centre distance was constructed from the same radius pair used to evaluate the geometry. The simplified relation therefore recovers $r_s = 0.1016 \text{ m}$ exactly at that calibration point, giving

$$R_g = 2.806033, \quad R_{\text{eff}} = 2.942286.$$

At maximum shift, the exact compatibility relation with $C = C_{\text{exact}}$ gives

$$r_{s,\text{exact}} = 0.066\,196\,036 \text{ m},$$

whereas applying Equation (1.6) with the centre distance that was itself inferred from the small-angle approximation gives

$$r_{s,\text{approx}} = 0.056\,255\,598 \text{ m}.$$

The resulting ratios are summarized in Table 1.2.

Configuration	Exact	Approximate	Relative error
R_g , zero shift	2.806033	2.806033	0
R_{eff} , zero shift	2.942286	2.942286	0
R_g , maximum shift	0.860726	0.731474	15.02%
R_{eff} , maximum shift	0.855970	0.722303	15.62%

Table 1.2: Propagation of the centre-distance approximation through the current reference geometry. The approximate column uses C_{approx} from Equation (1.3).

The large maximum-shift discrepancy is therefore not simply the local error of dropping the wrap-angle term at that operating point. It is mainly the consequence of first using the simplified relation to infer the wrong installed centre distance and then carrying that distance through the rest of the shift path.

1.3 If the True Centre Distance Is Already Known

That distinction can be checked directly. If the true installed centre distance C_{exact} is supplied to the simplified relation, rather than inferred from it, then at maximum shift

$$r_{s,\text{approx}|C_{\text{exact}}} = 0.066\,478\,830 \text{ m},$$

and the corresponding cord-line ratio is

$$R_{\text{eff,approx}}|_{C_{\text{exact}}} = 0.859772.$$

Relative to the exact value $R_{\text{eff}} = 0.855970$, the error is only

$$0.444\%.$$

For the present geometry, then, the strongest warning concerns using the small-angle relation to *identify the installed geometry* from the reference pulley radii and belt length. Once the true centre distance is known, the same reduced relation is much closer over the tested maximum-shift point.

The CINDER formulation avoids both distinctions by retaining the exact compatibility relation throughout. The numerical values in this chapter can be regenerated with the companion script `cvt_geometry_note_calculations.py`.

Chapter 2

Geometric CVT Ratio Rate, Acceleration, and Design Maps

The CINDER geometry formulation determines the individual pulley radii and their rates and accelerations from the shared shift motion. Those results can also be combined to describe how the available geometric ratio changes along the compatible pulley path.

Everything in this chapter inherits the reduced geometry used by CINDER: fixed belt length, prescribed circular wraps and straight spans, common sheave half-angle, and the compatible one-coordinate pulley motion. The resulting relations and plots are therefore design tools for that reduced geometry, not general laws for every CVT construction.

Define the outer-surface geometric ratio as

$$R_g = \frac{r_s}{r_p}. \quad (2.1)$$

The main formulation supplies the required pulley-radius rates in Section 3.1.4 and accelerations in Section 3.1.5. The sections below combine those relations and explore their geometric consequences.

2.0.1 Rate of Change of the Geometric CVT Ratio

Both the numerator and denominator in Equation (2.1) can change with time. Using the quotient rule,

$$\begin{aligned} \dot{R}_g &= \frac{d}{dt} \left(\frac{r_s}{r_p} \right) \\ &= \frac{\dot{r}_s r_p - r_s \dot{r}_p}{r_p^2} \\ &= \frac{\dot{r}_s}{r_p} - \frac{r_s}{r_p^2} \dot{r}_p. \end{aligned} \quad (2.2)$$

The secondary-radius result in Equation 3.1.57 supplies the coupled secondary motion. Within the active-shift branch,

$$\dot{r}_s = -\frac{\phi_p}{\phi_s} \dot{r}_p. \quad (2.3)$$

Substituting this into Equation (2.2) gives

$$\begin{aligned}
\dot{R}_g &= \frac{1}{r_p} \left(-\frac{\phi_p}{\phi_s} \dot{r}_p \right) - \frac{r_s}{r_p^2} \dot{r}_p \\
&= -\frac{\dot{r}_p}{r_p} \left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p} \right) \\
&= -\frac{\dot{r}_p}{r_p} \left(\frac{\phi_p}{\phi_s} + R_g \right).
\end{aligned} \tag{2.4}$$

Finally, Equation 3.1.36 gives $\dot{r}_p = \dot{s}/(2 \tan \beta)$ in the active-shift branch. Therefore,

Rate of Change of Geometric CVT Ratio

(2.5)

$$\dot{R}_g = \begin{cases} 0, & 0 \leq s < s_{dz}, \\ -\frac{\dot{s}}{2 \tan \beta r_p} \left(\frac{\phi_p}{\phi_s} + R_g \right), & s_{dz} < s \leq s_{max}. \end{cases}$$

For an upshift, $\dot{s} > 0$, so the available geometric reduction decreases as the belt moves outward on the primary pulley.

2.0.2 Geometry Families in the Radius Plane

The preceding result gives the local rate at which the geometric ratio changes once the pulley geometry is known. To interpret that result in terms of actual CVT design choices, it is useful to first ask which pulley-radius combinations are available in the first place.

Recall once more that the geometric CVT ratio is

$$R_g = \frac{r_s}{r_p}. \tag{2.6}$$

Thus, every point in the (r_p, r_s) plane corresponds to one geometric ratio, and every constant-ratio contour is a line of geometrically similar pulley states. This view is helpful because it separates three different design questions that are easy to blur together:

1. What geometric ratio does a given radius pair provide?
2. Which radius pairs are compatible with a selected belt and installed geometry?
3. How does the chosen compatible path influence the way ratio changes as the CVT shifts?

The first question is answered by the constant- R_g contours. The other two come from the belt-length compatibility relation. Once one of the global geometric quantities—belt length L_b or center-to-center distance C —is fixed, the compatibility relation defines a family of allowable radius pairs in the same plane.

Figure 2.1 compares these two views directly. The left panel fixes the center distance and varies the belt length, while the right panel fixes the belt length and varies the center distance. In both panels, the solid contours are constant geometric ratio. The dashed families are the geometrically compatible radius pairs selected by the remaining design choice.

Matched comparison of CVT geometry families in the radius plane

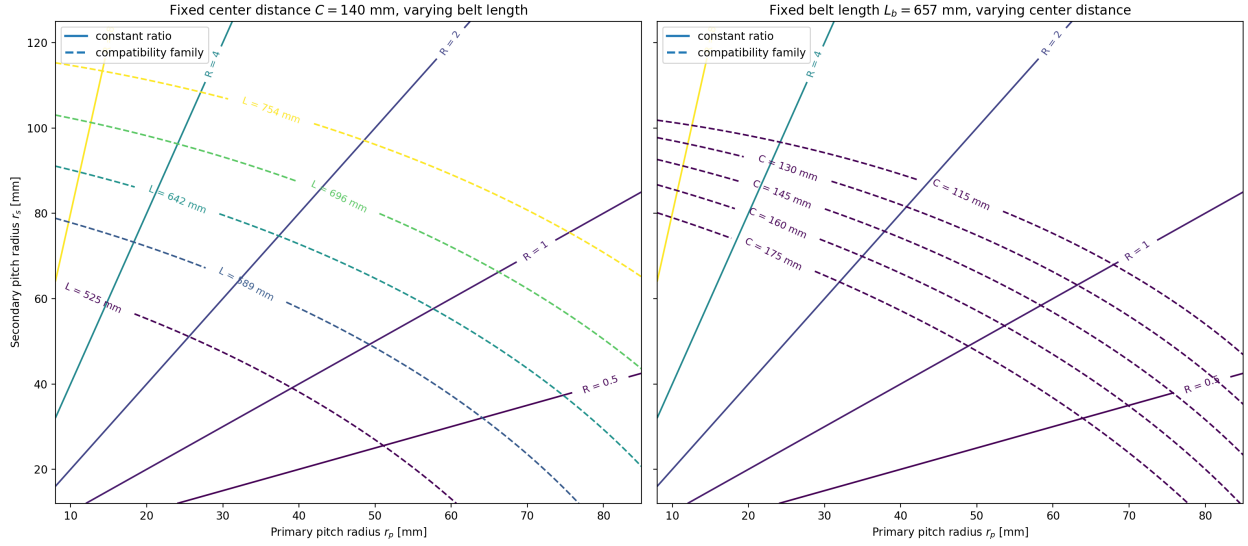


Figure 2.1: Matched comparison of geometry families in the pulley-radius plane. In both panels, the solid contours are constant geometric ratio $R_g = r_s/r_p$. In (a), the center-to-center distance is fixed and the dashed families correspond to different belt lengths. In (b), the belt length is fixed and the dashed families correspond to different center-to-center distances.

This figure makes two points especially clear.

First, meeting a target ratio aperture is not only a matter of choosing two endpoint ratios. The selected belt and installed geometry determine which radius pairs are even reachable. A longer or shorter belt shifts the compatible family through the plane, and so does a change in center distance. Since the constant-ratio lines are not parallel to those compatibility families, changing L_b or C changes not only the reachable endpoints, but also the geometric route between them.

Second, the figure makes the roles of belt selection and packaging easier to distinguish. The belt length is largely a component-selection choice, while the center distance is largely a packaging choice. Both appear in the same compatibility relation, but they move the admissible family in different ways. This is why a target ratio range cannot be discussed cleanly without also specifying either the chosen belt or the available center-to-center distance.

In this sense, Figure 2.1 is the static geometric view of the design problem. It shows what ratio states are available. The next question is dynamic in a geometric sense: once the CVT moves along one of those compatible paths, how much ratio change does a small increment of axial travel produce?

2.0.3 Exploring the Local Ratio-Change Map

The expression for $\dot{R}_g$ gives the instantaneous rate of change of the geometric ratio. To compare different parts of the geometry without committing to a particular time history, it is useful to divide out the shift speed and look instead at the local ratio change produced by a small increment of axial travel:

$$\frac{dR_g}{ds} = -\frac{1}{2 \tan \beta r_p} \left(\frac{\phi_p}{\phi_s} + R_g \right). \quad (2.7)$$

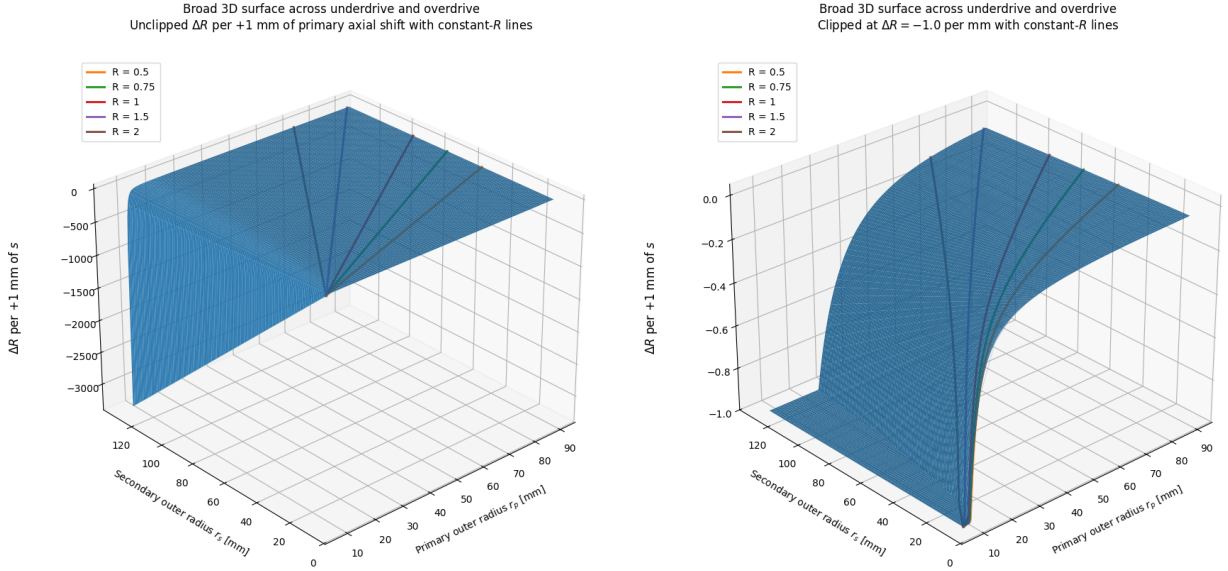
For a small positive increment Δs ,

$$\Delta R_g \approx \frac{dR_g}{ds} \Delta s. \quad (2.8)$$

The plots below take $\Delta s = 1$ mm, so the displayed quantity may be read as the local change in geometric ratio caused by an additional 1 mm of primary closure. The negative sign simply reflects the usual upshift direction: increasing s reduces the geometric reduction ratio.

A sensitivity field over the radius plane The first view is a broad surface over the radius plane. Here, r_p and r_s are treated as independent coordinates so that every part of the geometry can be inspected. A real CVT does not wander arbitrarily across this surface; its fixed belt length and center distance constrain it to one belt-length-compatible path. Even so, the full surface is valuable because it shows where the geometry itself makes ratio change especially strong or weak.

Figure 2.2 shows two versions of this surface. The unclipped view exposes the dominant trend: the local ratio change becomes much larger in magnitude as the primary radius becomes small. This follows directly from the explicit $1/r_p$ factor in Equation (2.7), and is reinforced by the fact that the geometric ratio $R_g = r_s/r_p$ is also largest in that same region. The clipped view hides the most extreme values so that the otherwise compressed mid-range curvature can be seen more clearly. The orange curve marks the $R_g = 1$ locus. It is a convenient reference, but not a special boundary in the local ratio-change law itself.



(a) Broad radius domain. The full scale exposes the rapidly increasing magnitude of local ratio change near small r_p .

(b) The same domain with values below -1 per 1 mm clipped, exposing the curvature through the mid-range.

Figure 2.2: Local geometric-ratio change for a 1 mm increase in primary axial closure across a broad pulley-radius domain. The orange curve marks the $R_g = 1$ locus.

How belt choice and installed geometry select a path The broad surface is useful, but a particular CVT design only experiences the part of that field lying along its own compatible geometry path. This is where the earlier radius-plane view becomes directly relevant. Once a

belt length and center distance are fixed, the CVT is constrained to one compatibility curve in the (r_p, r_s) plane. As the primary sheave closes, the operating point moves along that curve and samples the local ratio-change field point by point.

Figure 2.3 overlays three representative belt-length-compatible paths on the local ratio-change surface. Each path corresponds to a different belt-length/reference-geometry choice. The figure shows that two designs with similar overall ratio aperture can still distribute their ratio change very differently over the available shift travel. One path may enter the steep small- r_p region early, concentrating a large amount of ratio change into a short portion of the stroke. Another may spend more of the stroke in the flatter region, distributing the same overall change more gradually.

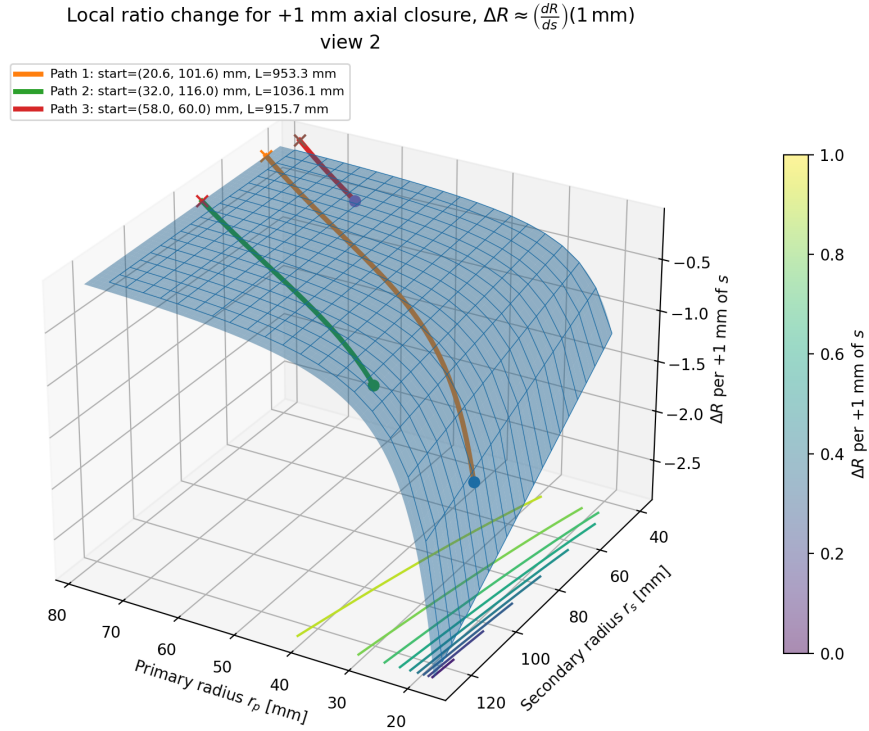


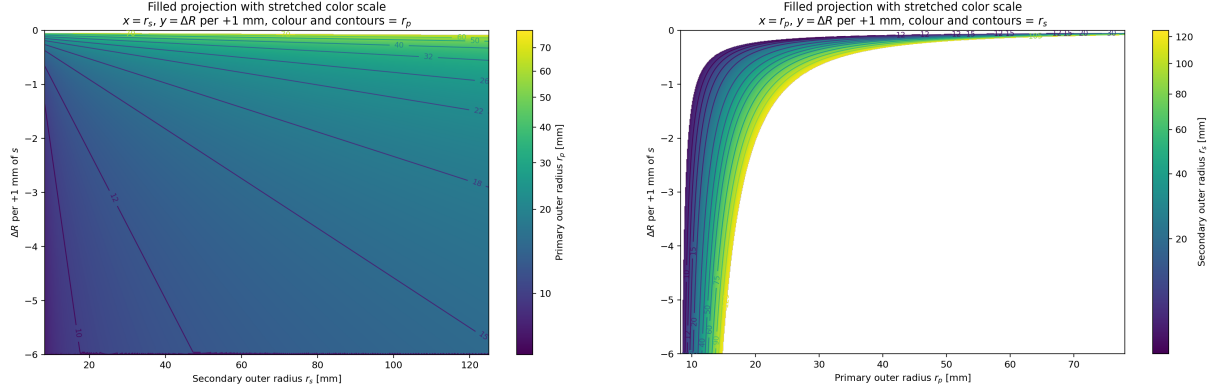
Figure 2.3: Representative fixed-belt-length paths through the local ratio-change surface. Each coloured curve is a geometrically compatible pulley-radius path for a different belt-length/reference-geometry choice. The surface value gives the local change in geometric ratio associated with an additional 1 mm of primary closure at that point.

This is the key design value of the map: it does not merely say what overall ratio range is available. It shows *where along the shift stroke* that ratio change is spent.

Separating the roles of the two pulley radii The two projections in Figure 2.4 help isolate the individual effects of r_p and r_s . In Figure 2.4(a), the horizontal axis is the secondary radius and the contours identify the primary radius. For a fixed primary radius, increasing r_s increases the geometric ratio and makes the local ratio change more negative. The effect is visible, but the contour spacing shows that it remains secondary to the influence of the primary radius.

Figure 2.4(b) makes the primary-radius effect more explicit. Every secondary-radius contour bends sharply downward as r_p approaches its lower limit, then flattens at larger radius. This is the

visual form of the $1/r_p$ dependence in Equation (2.7). Thus, the small-primary-radius region is not only where the geometric reduction is large; it is also where the ratio becomes most sensitive to additional axial motion.



(a) Local ratio change against secondary radius. Colour and contours identify the primary radius.

(b) Local ratio change against primary radius. Colour and contours identify the secondary radius.

Figure 2.4: Two projections of the local ratio-change surface for a 1 mm primary closure. The projections separate the contributions of the two radii and show the dominant sensitivity to small primary radius.

Design interpretation Taken together, Figure 2.1 and Figure 2.2–Figure 2.4 give a more complete geometric design picture than endpoint ratios alone.

The radius-plane view answers the question of feasibility. It makes clear how a target ratio aperture depends on the selected belt length and the available center-to-center distance. A longer or shorter belt can move the admissible family enough to change both the reachable ratio range and the portion of the plane through which the CVT shifts. Likewise, a packaging change in center distance can move the same belt into a very different family of compatible radius pairs.

The ratio-change view answers the question of distribution. Once a design selects its belt and installed geometry, the compatible path through the plane determines where ratio change is concentrated. A path that reaches very small r_p can produce large ratio change in little axial travel, which may be attractive when launch or climbing performance demands strong reduction with limited shift stroke. A path that stays farther from that corner distributes ratio change more gently and may be easier to control or to package within a desired sheave travel.

This does *not* mean that the smallest possible r_p is always best. The present plots are geometric only. They show the leverage of the geometry, but not its cost. Experimental work by Chen et al. [1998] showed that, for their rubber-belt CVT, the highest transmission efficiency occurred near a 1:1 speed ratio, reminding us that the most attractive geometric reduction state is not automatically the most efficient operating state. More directly, Childs and Cowburn [1987] identified additional loss penalties at small pulley radii in V-belt drives, with belt-carass warping and shearing proposed as important contributors. In other words, the same small- r_p region that offers strong geometric leverage is also the region where the belt is bent and seated most severely.

For that reason, these plots are best used as an early design lens rather than a final design verdict. They help screen combinations of belt length, center distance, and pulley radii by showing not only how much ratio range is available, but how that range is distributed over the stroke.

Later traction, loss, and dynamic analyses can then be used to judge whether the geometrically attractive part of the design space is physically supportable [Gerbert, 1996, Bertini et al., 2014, Messick, 2018].

2.0.4 Acceleration of the Geometric CVT Ratio

The ratio rate can change for two reasons. The primary sheave may accelerate, and the pulley geometry continues to change as the belt moves. Differentiate Equation (2.2) term by term to capture both effects.

For the first term, $\dot{r}_s/r_p$, use the quotient rule:

$$\frac{d}{dt} \left(\frac{\dot{r}_s}{r_p} \right) = \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s \dot{r}_p}{r_p^2}. \quad (2.9)$$

For the second term, $r_s \dot{r}_p / r_p^2$, all three factors may change. Using the product rule,

$$\frac{d}{dt} \left(\frac{r_s \dot{r}_p}{r_p^2} \right) = \frac{\dot{r}_s \dot{r}_p}{r_p^2} + \frac{r_s \ddot{r}_p}{r_p^2} - \frac{2r_s \dot{r}_p^2}{r_p^3}. \quad (2.10)$$

Subtracting Equation (2.10) from Equation (2.9) gives

$$\begin{aligned} \ddot{R}_g &= \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s \dot{r}_p}{r_p^2} - \left[\frac{\dot{r}_s \dot{r}_p}{r_p^2} + \frac{r_s \ddot{r}_p}{r_p^2} - \frac{2r_s \dot{r}_p^2}{r_p^3} \right] \\ &= \frac{\ddot{r}_s}{r_p} - \frac{2\dot{r}_s \dot{r}_p}{r_p^2} - \frac{r_s \ddot{r}_p}{r_p^2} + \frac{2r_s \dot{r}_p^2}{r_p^3}. \end{aligned} \quad (2.11)$$

The current formulation gives the corresponding secondary rate and acceleration in Equation 3.1.57 and Equation 3.1.83. On the engaged branch, those results can be rewritten in terms of the primary radius motion as

$$\dot{r}_s = -\frac{\phi_p}{\phi_s} \dot{r}_p, \quad (2.12)$$

$$\ddot{r}_s = -\frac{\phi_p}{\phi_s} \ddot{r}_p - \frac{8\pi^2}{\phi_s^3 \sqrt{C^2 - (r_s - r_p)^2}} \dot{r}_p^2. \quad (2.13)$$

Substituting these into Equation (2.11) gives

$$\begin{aligned} \ddot{R}_g &= \frac{1}{r_p} \left[-\frac{\phi_p}{\phi_s} \ddot{r}_p - \frac{8\pi^2}{\phi_s^3 \sqrt{C^2 - (r_s - r_p)^2}} \dot{r}_p^2 \right] - \frac{2}{r_p^2} \left[-\frac{\phi_p}{\phi_s} \dot{r}_p \right] \dot{r}_p \\ &\quad - \frac{r_s \ddot{r}_p}{r_p^2} + \frac{2r_s \dot{r}_p^2}{r_p^3} \\ &= -\frac{1}{r_p} \left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p} \right) \ddot{r}_p + \left[\frac{2}{r_p^2} \left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p} \right) - \frac{8\pi^2}{\phi_s^3 r_p \sqrt{C^2 - (r_s - r_p)^2}} \right] \dot{r}_p^2 \\ &= -\frac{1}{r_p} \left(\frac{\phi_p}{\phi_s} + R_g \right) \ddot{r}_p + \left[\frac{2}{r_p^2} \left(\frac{\phi_p}{\phi_s} + R_g \right) - \frac{8\pi^2}{\phi_s^3 r_p \sqrt{C^2 - (r_s - r_p)^2}} \right] \dot{r}_p^2. \end{aligned} \quad (2.14)$$

Finally, use the primary radius rate and acceleration from Equation 3.1.36 and Equation 3.1.66:

Acceleration of Geometric CVT Ratio

(2.15)

$$\ddot{R}_g = \begin{cases} 0, & 0 \leq s < s_{dz}, \\ -\frac{\ddot{s}}{2 \tan \beta r_p} \left(\frac{\phi_p}{\phi_s} + R_g \right) + \frac{\dot{s}^2}{2 \tan^2 \beta r_p^2} \left(\frac{\phi_p}{\phi_s} + R_g \right) - \frac{2\pi^2 \dot{s}^2}{\phi_s^3 r_p \tan^2 \beta \sqrt{C^2 - (r_s - r_p)^2}}, & s_{dz} < s \leq s_{\max}. \end{cases}$$

The ratio acceleration has two sources. The term proportional to $\ddot{s}$ is the direct conversion of primary-sheave acceleration into ratio acceleration. Its magnitude is scaled by the local ratio sensitivity dR_g/ds , so the same axial acceleration produces a larger ratio response where the geometric ratio changes rapidly with shift.

The terms proportional to $\dot{s}^2$ arise even when the primary sheave moves at constant axial speed. They reflect the curvature of the geometric relation $R_g(s)$: as the belt moves through the sheaves, the ratio change produced by each additional increment of axial travel need not remain constant. In the usual upshift direction, the large ratio sensitivity associated with small primary radius progressively weakens as the primary radius grows. The ratio therefore often changes rapidly at the beginning of the shift and then eases off, even without a change in axial shift speed.

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